Data Details in SkillTab-Bench

Data Source of SkillTab-Bench

The tables used for benchmarking are sourced from both publicly available online resources and commercially purchased datasets. The online resources include municipal open data platforms, the official website of the National Bureau of Statistics, industry association portals, and open table datasets, with specific data sources listed in Table 7. Commercially purchased data consists of open-license tables and industry reports obtained from professional data service providers. Additionally, our proprietary collection includes anonymized tables derived from real-world industrial applications.

Sources	Websites
Open-source data platforms	
World Bank Group	https://datacatalog.worldbank.org/
National Bureau of Statistics of China	https://www.stats.gov.cn/sj/
Kaggle	https://www.kaggle.com/datasets
China Association of Automobile Manufacturers	http://www.caam.org.cn/
Beijing Public Data Open Platform	https://data.beijing.gov.cn/
The United States Government's Open Data Site	https://catalog.data.gov/dataset
China Securities Regulatory Commission Data Platform	http://www.csrc.gov.cn/csrc/tjsj/index.shtml
Shanghai Public Data Open Platform	https://data.sh.gov.cn/view/data-resource/index.html
CelesTrak	https://celestrak.org/
Tabular datasets	
MiMoTable (Li et al., 2024)	https://github.com/jasonNLP/MiMoTable
WikiTQ (Pasupat & Liang, 2015)	https://github.com/ppasupat/WikiTableQuestions

Detailed Definitions and Examples of DAG Structures

This paper introduces six DAG structures to describe the dependencies and interaction patterns between information points in complex multi-table tasks. Each DAG represents a distinct type of reasoning process and task execution path, aiding in the clarification of multi-turn dialogue flow and providing a structured framework for evaluating model reasoning capabilities. The following sections present the definitions, applicable scenarios, and specific examples of these topologies, enabling readers to better understand how to design tasks using these structures and apply them to multi-table reasoning tasks. The DAG definitions are as follows:

1. Chain

- Definition: Information points strictly depend on each other in a linear sequence, with a progressive relationship, where the next question relies on the context of the previous one.
- Applicable Scenarios: Suitable for tasks that require step-by-step analysis, where each step builds on the result of the previous one.
- Example:
  A = Top five provinces by GDP
  B = Which of these provinces have national-level AI demonstration zones?
  C = Which provincial capitals have the highest number of AI companies?
  D = Which AI company has the highest per capita output?
- Diagram:
  $A \rightarrow B \rightarrow C \rightarrow D$

2. Tree

- Definition: Starting from a root node, the structure branches downward, containing at least two levels; sub-branches can further refine the analysis, but paths within each branch remain independent and chain-like.
- Applicable Scenarios: Suitable for obtaining overall information first, followed by a layered breakdown into specific details, forming a hierarchical expansion.
- Example:
  A = Major food crops in China (assumed to be rice and wheat)
  B = Major rice-producing provinces (assumed to be Jiangxi and Hunan)
  C = Major wheat-producing provinces (assumed to be Henan)
  D = Rice yield in Hunan province
  E = Rice planting area in Jiangxi province
  F = Total wheat production in Henan
- Diagram:
  $A \rightarrow B$, $A \rightarrow C$, $B \rightarrow D$, $B \rightarrow E$, $C \rightarrow F$

3. Fork-join

- Definition: Multiple information points are independently retrieved in parallel (without dependencies), and later combined into a single comprehensive conclusion
- Applicable Scenarios: Suitable for analyzing the same theme from multiple independent angles, then merging the results into a final conclusion.
- Example:
  A = AI industry scale in Beijing in 2024
  B = AI industry scale in Shanghai in 2024
  C = AI industry scale in Shenzhen in 2024
  D = Combine data from the three cities to recommend the most suitable city for setting up an R&D center
- Diagram:
  $A \rightarrow D$, $B \rightarrow D$, $C \rightarrow D$

4. Star

- Definition: A central information point is queried in parallel across multiple orthogonal dimensions; all sub-questions depend only on the central point, and no further expansion occurs within each sub-path.
- Applicable Scenarios: Suitable for asking multiple questions about the same table or entity from different attribute dimensions, without delving deeper into any one dimension.
- Example:
  A: Query the record with order ID ORD1001
  B: What is the customer's name?
  C: When was the order placed?
  D: What is the total amount of the order?
- Diagram:
  $A \rightarrow B$, $A \rightarrow C$, $A \rightarrow D$

5. Mesh

- Definition: After multiple branches expand, subsequent queries can freely jump back to historical nodes of different branches, forming cross-path references.
- Applicable Scenarios: Suitable for non-continuous, cross-branch context referencing, simulating the human ability to freely jump between topics.
- Example:
  A: Query total sales data for all categories in Q1 2024
  B: Sales of electronics
  C: Sales of home appliances?
  D: Which category of electronics has the highest profit margin?
  E: What is the percentage of air conditioners in the home appliance sales mentioned in Q3? (At this point, the Q3 branch has ended, and Q5 actively jumps back to reference Q3)
- Diagram:
  $A \rightarrow B$, $A \rightarrow C$, $B \rightarrow D$, $D \rightarrow E$,
  E $\dashrightarrow$ Cross-branch backtrack C

Task Templates and Task Design Details

In the design of multi-table reasoning multi-turn dialogue tasks, each task is automatically generated based on the table information from the data folder and employs an appropriate DAG to organize the reasoning process. This process includes selecting task templates, randomly choosing topologies, and generating candidate tasks with topological structures. The following is the task generation prompt template:

You are an expert in constructing multi-turn dialogues, specializing in extracting valuable information from complex data to help users design tasks suitable for multi-turn reasoning. When designing tasks with topological structures, you should follow these steps:
1. Randomly select 5 task templates:
Randomly choose 5 task templates from the task template list.
2. Choose a topology based on the task:
For each task template, randomly select 2 to 4 appropriate topologies and generate tasks suitable for multi-turn reasoning based on the provided table data.
3. Output multi-turn reasoning tasks:
Generate multi-turn reasoning tasks based on the task templates and topologies, ensuring each task clearly defines the scenario, purpose, content, and reasoning topology.
Each task should have a clear scenario, purpose, analysis content, and the appropriate topology to ensure a well-defined reasoning path in the multi-turn dialogue.

Key Concepts
Task (Core Task, Main Query):
The core goal of the task or the analytical need the user seeks to fulfill, typically involving the task's scenario, analytical objective, and required data.
Inference Topology:
The data path designed to complete the task, determining the sequence of sub-questions during the analysis process. By selecting the appropriate topology, it helps organize the task's multi-turn reasoning.

Scenario Design Principles:
Task Background:
Clarify the background, analytical objective, and scope of the task. Each task description should include background, purpose, and analysis scope.
Design Thinking:
Follow the topological structure design thinking and define the logical flow of each dialogue round. Every step of the task should clearly demonstrate the data queries and reasoning process.

Input Data:
Table Data Summary (Multiple tables are separated by "——")
{table_summaries}

Task Templates:
Please randomly select 5 task templates from the following list and analyze them with the provided table data:
1.**Comparative Analysis**
Task Description: Compare the differences and similarities across multiple entities, objects, or time periods.
Example Queries:
-Compare the sales growth of different regions between 2020 and 2021, and analyze the discrepancies.
-Compare the market performance of multiple products across different time periods, exploring their similarities and differences.
2.**Trend Analysis**
Task Description: Analyze the trend, growth or decline patterns, and evolution of time-series data.
Example Queries:
-Analyze the annual sales trend of a particular industry over the past five years and forecast the upcoming months.
-Study the yearly temperature trends of a city and uncover its long-term growth or decline patterns.

3.**Distribution Analysis**
Task Description: Analyze the distribution patterns, structural proportions, and ratio relationships of the data.
Example Queries:
-Analyze the proportion of different age groups using a particular service and identify the primary user group.
-Examine the income distribution across regions and assess income inequality.
4.**Causal Analysis**
Task Description: Investigate the causes, influencing factors, drivers, and causal relationships.
Example Queries:
-Analyze the causal relationship between advertising expenditure and sales growth, identifying key influencing factors.
-Examine the relationship between educational investment and student performance, exploring potential drivers.
5.**Evaluation and Diagnosis**
Task Description: Evaluate a situation, diagnose problems, identify risks, and detect anomalies.
Example Queries:
-Assess the financial status of a company and analyze potential financial risks.
-Detect anomalies in product sales data and identify possible causes.
6.**Ranking and Filtering**
Task Description: Rank objects, filter top/bottom values, and identify extremes.
Example Queries:
-Rank products based on sales volume and identify the top 10 highest-selling items from the past year.
-Filter the 5 regions with the lowest sales in 2019 and analyze the underlying causes.
7.**Aggregation and Statistics**
Task Description: Aggregate and compute statistics such as sums, averages, counts, etc.
Example Queries:
-Calculate the average annual income of a city over the past three years and analyze the trends.
-Summarize product sales data to identify the best-selling products and the regions with the highest sales.
8.**Correlation Analysis**
Task Description: Analyze correlations between variables, association rules, and relationships.
Example Queries:
-Analyze the correlation between advertising spending and sales, identifying the optimal balance between investment and returns.
-Investigate the purchasing correlation between different products and explore potential association rules.
9.**Prediction and Inference**
Task Description: Predict future trends, infer outcomes, and make projections based on existing data.
Example Queries:
-Predict the sales volume of a product in the upcoming months, based on historical sales data and market trends.
-Forecast the rainfall patterns for the next quarter based on weather data from previous years.
10.**Comprehensive Reporting**
Task Description: Generate a comprehensive analysis report involving multidimensional analysis.
Example Queries:
-Write a report on the current state and future trends of a particular industry, incorporating market data, policy environment, and competitive landscape.
-Provide a cross-sector report that compares various regions, product categories, and time periods, offering conclusive recommendations.

Topology Structure Annotation:
Based on the table data and task templates, randomly select 2 to 4 suitable topologies and annotate them for the task:
{topology_definitions}

Output Format:
Based on the task template, topology structure, and table data, generate tasks suitable for multi-turn reasoning, and annotate the appropriate topological structures.

```
[
  {
    "task": "Analyze sales trends across different regions over the past five years, identify significant growth or decline, and predict market trends for the upcoming months.",
    "design_thinking": "Following the comparative analysis topology, first extract sales data from Table 1, then perform an annual comparison, followed by identifying fluctuation patterns across regions. Finally, combine multidimensional analysis with trend analysis to form sales predictions.",
    "type": "Tree"
  },
  ...
]
```

Relationship between DAG Structures and Task Types

By analyzing the sample proportions of each DAG structure and the distribution of task types, we quantify the association strength between dialogue topologies and task types using the Lift metric. The formula for calculating Lift is as follows:

$$\mathrm{Lift}(T, \tau) = \frac{P(\tau|T)}{P(\tau)} = \frac{\frac{N_{T,\tau}}{N_T}}{\frac{N_\tau}{N}}$$

Where T represents the DAG structure type, τ represents the task type, NT,τ denotes the number of samples belonging to both DAG T and task type τ , NT is the total number of samples for DAG T , Nτ is the total number of samples for task type τ , and N is the total number of samples (1310). We adopt a threshold of 1.2 to filter weak associations and improve interpretability; accordingly, only connections with a Lift value greater than 1.2 are visualized.

The results show that tree and fork-join topologies have the highest sample proportions, at 32.9% and 27.6%, respectively, followed by chain DAG at 24.6%. Regarding task types, comparative analysis is the most common, occurring 266 times, followed by comprehensive reporting and trend analysis, with 114 and 83 occurrences, respectively. The strong association analysis reveals that chain DAG is most strongly associated with trend analysis, causal analysis, and correlation analysis (Lift > 1.2), while mesh DAG shows a significant preference for comparative analysis (Lift = 1.51). Star and mixed topologies demonstrate a marked preference for distribution analysis, with Lift values of 3.47 and 3.58, respectively. Notably, loop-back DAG is almost exclusively associated with comparative analysis (Lift = 2.22). Overall, large-sample topologies (e.g., tree and fork-join) exhibit more diverse task type associations, while small-sample topologies show a strong preference for a few specific task types.

Dialogue Synthesis

In this study, we propose a DAG-aware dialogue synthesis method for generating dialogues in multi-turn reasoning tasks. The process encompasses the steps involved in multi-turn dialogue generation and integrates quality control strategies to ensure that the generated dialogues meet accuracy and naturalness standards. We adopt a three-stage synthesis method to generate multi-turn dialogues through the following three-phase prompts:

GENERATION_DIALOGUE_PROMPT_STAGE1

Task Background
The goal is to generate multi-turn dialogues based on Q&A data from a complex task. It is important to note the fundamental difference between Q&A and multi-turn dialogue: while Q&A provides direct answers to specific, clear questions, multi-turn dialogue involves progressively deeper questioning around the user's conversational intent.

In designing multi-turn dialogues, please follow these steps:
1. Define the Dialogue Task (Task): Clarify the overall goal of the user at the start of the dialogue.
2. Design the Information Retrieval Path: Construct a logically structured information retrieval strategy around the topic, specifying the data required at each step and the corresponding answers.
3. Generate the Dialogue Content: Based on the information retrieval path designed above, generate natural multi-turn dialogue.

Input Data:
Dialogue Task (Task)
{task}

Table Summary (Tables are separated by "——")
{table_summary}

Task Objective
You are required to design a multi-turn dialogue plan based on the provided inputs, referencing the given `tupu_type`. The plan should include three components:

1. Core Task (Task): Refine and rephrase the Dialogue Task (Task) into a natural language description of the user's overall goal at the beginning of the dialogue. The background context and overall purpose should be outlined first, followed by specific details of the information or scope to be queried, while ensuring it aligns with the original task, given topology, and table summary.
2. Information Points List (checkout_list): Decompose the core task into several information retrieval steps, each representing data the user intends to retrieve.
- From the User's Perspective: Describe the information to be obtained from the user's point of view.
- Topological Structure: Organize the information retrieval according to the specified topology, simulating the user's thought process.
- Coverage: Ensure the structure is maintained while covering all the information from the original answer. Information may be omitted to maintain the topological structure.
- Validity: Each "info_item" and "answer" must be derivable from the provided Table Summary.

3. Topology: Define the logical dependencies between the information points.
Ensure the information checkpoints strictly adhere to the chosen topology.

Topology
{tupu_type}

Output Format
After reflecting on the above, provide your final answer in JSON format:

```
[
  {
    "task": "Core task description...",
    "design_thinking": "Topology design and construction process...",
    "checkout_list": [
    {
      "idx": 0,
      "info_item": "Description of information point 1...",
      "answer": "Expected answer for this information point..."
    },
    {
      "idx": 1,
      "info_item": "Description of information point 2...",
      "answer": "Expected answer for this information point..."
    },
    ...
```

```
    ],
    "type": "Topology type",
    "graph": "mermaid graph \n ...\n"
  },
  ...
]
```

GENERATION_DIALOGUE_PROMPT_STAGE2

```
# Task Background
The goal is to generate multi-turn dialogues based on Q&A data from a complex task. It is important to distinguish between Q&A and multi-turn dialogue: while Q&A involves direct responses to individual, clear questions, multi-turn dialogue progressively explores related questions surrounding the user's conversational intent.

In designing multi-turn dialogues, please follow these steps:
1. Define the Dialogue Task (Task): Clarify the user's overall goal at the beginning of the dialogue.
2. Design the Information Retrieval Path: Construct a logically structured information retrieval plan, specifying the data needed at each step and the corresponding answers.
3. Generate the Dialogue Content: Generate natural multi-turn dialogue based on the designed information retrieval path.

# Task Requirements
The topology structure path has already been generated. Please proceed with refining the checkpoints while maintaining the topology. This can involve reordering adjacent checkpoints or incorporating more information from the original answers.
- Compress the checkpoints to 2-6: If checkpoints are too long, keep the topology intact while compressing the checkpoints.
- Diversity of Questions: While restructuring, feel free to rephrase some checkpoints as more varied questions, including filters, sorting, and aggregation.
- Increased Difficulty: Some checkpoints can be made more complex by introducing composite indicators, numerical reasoning problems, multi-table reasoning, or comparative analysis.

Finally, optimize the task description, design thinking, and graph information based on the revised checkpoint list.

# Input Data:
## Dialogue Task (Task)
{task}

## Existing Design and Checkpoints
{multi_dialogue_design}

# Output Format
Provide your final answer in JSON format:

[
  {
    "task": "Core task description...",
    "design_thinking": "Topology design and construction process...",
    "checkout_list": [
    {
      "idx": 0,
        "info_item": "Description of information point 1...",
      "answer": "Expected answer for this information point..."
    },
    {
      "idx": 1,
      "info_item": "Description of information point 2...",
      "answer": "Expected answer for this information point..."
    },
    ...
    ],
```

```
    "type": "Topology type",
    "graph": "mermaid graph LR\n ...\n"
  },
  ...
]
```

GENERATION_DIALOGUE_PROMPT_STAGE3

```
# Task Background
The goal is to generate multi-turn dialogues based on Q&A data from a complex task. It is important to note the fundamental difference between Q&A and multi-turn dialogue: Q&A provides direct answers to specific, well-defined questions, whereas multi-turn dialogue involves progressively delving into related questions based on the user's conversational intent.

In designing multi-turn dialogues, please follow these steps:
1. Define the Dialogue Task (Task): Clarify the overall goal of the user at the start of the dialogue.
2. Design the Information Retrieval Path: Construct a logically structured approach to information retrieval, specifying the required data and corresponding answers at each step.
3. Generate the Dialogue Content: Generate multi-turn dialogue based on the designed information retrieval path.

# Task Requirements
The checkpoints are already designed, but I need you to fill in the following fields based on context:
1. Double-check the accuracy of each checkpoint's answer. If it is incorrect, revise it. Ensure that "info_item" and "answer" can be derived from the provided Table Summary.
2. Based on the table content, assign the associated table(s) to each checkpoint and add a "related_table" field with the table file name(s).
3. Add a conclusion field: Attempt to answer the user's core task based on all the checkpoints.

# Input Data:
## Dialogue Task (Task)
{task}

## Table Summary (Tables are separated by "——")
{table_summary}

## Existing Design and Checkpoints
{multi_dialogue_design}

# Output Format
Provide your final answer in JSON format:

[
  {
    "task": "Core task description...",
    "design_thinking": "Topology design and construction process...",
    "checkout_list": [
    {
      "idx": 0,
      "info_item": "Description of information point 1...",
      "answer": "Expected answer for this information point...",
      "related_table": ["Table file name 1", ...]
    },
    ...
    ],
    "conclusion": "Attempt to answer task",
  }
]
```

Data Augmentation

To enhance the overall fluency and naturalness of the generated multi-turn dialogues, we further improve them through coreference resolution, ellipsis restoration, and paraphrasing. This results in diversity in both structure and content, closely resembling real human user interactions. The prompts for the 3 types of data augmentation methods are as follows:

DATA_AUGMENTATION_PROMPT_V1

You are a professional large model for enhancing multi-turn dialogues. I would like you to help enhance the expression of sub-questions during a multi-turn dialogue process.

Task Background

Question answering (QA) and multi-turn dialogue have fundamental differences: QA is a direct response to a single, specific question, while multi-turn dialogue involves progressively asking related questions around the user's conversational intent. Therefore, multi-turn dialogue contains the following elements:

1. Defining the conversation topic (Task): The overall intent and objective of the user at the start of the conversation.
2. Designing the information retrieval path: The approach the user constructs around the topic to gather information, including each step's required information and corresponding answers.
3. Enhanced dialogue content: The actual questions and answers posed based on each information step mentioned above.

Task Requirements

The information retrieval path has already been defined. I would like you to enhance the expression of each information step's question based on this path. In this process, you cannot modify the semantics of each information step; instead, you should generate enhanced expressions that maintain the original meaning.

1. The questions you enhance should be as concise as possible, omitting intermediate indicators that are redundantly included, and only keeping non-redundant query indicators.

* For example: Calculate the production and export volumes of a product, then calculate the export ratio. You only need to ask to calculate the export ratio, as the production and export volumes will naturally be calculated when solving the export ratio problem.
2. For excessive indicators that cannot be combined, you need to think about whether they can be replaced by synthesized indicators that encapsulate the original multiple indicators. The synthesized indicator should semantically include the original indicators and be computable entirely from the original indicators.
3. For trend analysis or similar questions, consider whether they can be replaced with more specific indicators and questions to avoid ambiguity. Based on the enhanced expression above: Please also modify the answers to align with the enhanced question expressions.

Each sub-question's modified expression should be semantically equivalent to the original expression, and the modified answer should be derived entirely from the original answer.

Input Information

1. Multi-turn Dialogue Topic

{task}

2. Information Retrieval Approach

{design$_t$*hinking*}

3. Original Information Steps

{checkout_list}

Output Requirements

Please first analyze step by step and finally provide the answer in JSON format:

```
{{
  "checkout_list": [
    {{
      "id": Sub-question Number (idx),
      "info_item": "Enhanced Expression",
      "answer": "Answer to this expression",
    }},
    ...
```

```
  ]
}}
```

DATA_AUGMENTATION_PROMPT_V2

You are a professional large model for enhancing multi-turn dialogues. I would like you to help enhance the expression of sub-questions during a multi-turn dialogue process.

Task Background

Question answering (QA) and multi-turn dialogue have fundamental differences: QA is a direct response to a single, specific question, while multi-turn dialogue involves progressively asking related questions around the user's conversational intent. Therefore, multi-turn dialogue contains the following elements:

1. Defining the conversation topic (Task): The overall intent and objective of the user at the start of the conversation.
2. Designing the information retrieval path: The approach the user constructs around the topic to gather information, including each step's required information and corresponding answers.
3. Enhanced dialogue content: The actual questions and answers posed based on each information step mentioned above.

Task Requirements

The information retrieval path has already been defined. I would like you to enhance the expression of each information step's question based on this path.
1.**Manufacturing condition omission**:Please randomly select 1-2 questions in the multi-turn dialogue and intentionally omit critical limiting conditions (e.g., year, region, specific product model, etc.), making the question expression vague or incomplete.
-Purpose:This vagueness should force the table_agent to be unable to directly answer and instead prompt the user for clarification to confirm the specific conditions.
-Example:The original question is "What is the profit margin for the first quarter of 2023?" which is modified to "What is the profit margin?" (missing the time constraint).
2.**Keep other questions normal**:Except for the 1-2 vague questions selected above, the expressions of the other questions should remain semantically complete, clear, and logically consistent within the conversation.
3.**Natural language style**:Simulate a user who may express themselves imprecisely due to urgency or preset context.

Note: For the modified vague questions, please retain the **correct answer under complete conditions** (i.e., the answer the user originally intended to query with specific conditions).

Input Information

1. Multi-turn Dialogue Topic

{task}

2. Information Retrieval Approach

{design$_t hinking$}

3. Original Information Steps

{checkout_list}

Output Requirements

Please first analyze step by step and finally provide the answer in JSON format:

```
{{
  "checkout_list": [
    {{
      "id": Sub-question Number (idx),
      "info_item": "Enhanced Expression",
      "answer": "Answer to this expression",
    }},
    ...
  ]
}}
```

DATA_AUGMENTATION_PROMPT_V3

You are a professional large model for enhancing multi-turn dialogues. I would like you to help enhance the expression of sub-questions during a multi-turn dialogue process.

Task Background

Question answering (QA) and multi-turn dialogue have fundamental differences: QA is a direct response to a single, specific question, while multi-turn dialogue involves progressively asking related questions around the user's conversational intent. Therefore, multi-turn dialogue contains the following elements:

1. Defining the conversation topic (Task): The overall intent and objective of the user at the start of the conversation.
2. Designing the information retrieval path: The approach the user constructs around the topic to gather information, including each step's required information and corresponding answers.
3. Enhanced dialogue content: The actual questions and answers posed based on each information step mentioned above.

Task Requirements

The information retrieval path has already been defined. I would like you to enhance the expression of each information step's question based on this path. In this process, you cannot modify the semantics of each information step; instead, you should generate enhanced expressions that maintain the original meaning. 1. **Persona creation**: Please assume the persona of someone who speaks very concisely, with a direct and efficient questioning style, avoiding redundant embellishments.
2. **Introducing references and omissions**: In the sub-question expressions of the multi-turn dialogue, introduce references (e.g., "it," "that," "the former") and omissions (e.g., omitting subjects or known conditions) as much as possible to make the dialogue more aligned with natural human conversational habits.
3. **Increasing reasoning difficulty**: Through the use of references and omissions, increase the contextual dependence and reasoning complexity for the table_agent, forcing it to rely on previous dialogue information to accurately understand the intent of the current question.

Each sub-question's modified expression should be semantically equivalent to the original expression, and the modified answer should be derived entirely from the original answer.

Input Information

1. Multi-turn Dialogue Topic

{task}

2. Information Retrieval Approach

{design_thinking}

3. Original Information Steps

{checkout_list}

Output Requirements

Please first analyze step by step and finally provide the answer in JSON format:

```
{{
  "checkout_list": [
    {{
      "id": Sub-question Number (idx),
      "info_item": "Enhanced Expression",
      "answer": "Answer to this expression",
    }},
    ...
  ]
}}
```

Automated Evaluation

The automated evaluation tool performs a step-by-step assessment of each dialogue sample, primarily focusing on the correctness of sub-questions, table dependencies, and difficulty level. The evaluation process includes:

- Sub-question Level Evaluation: Each sub-question is evaluated for correctness, table dependency, and difficulty, ensuring that the question is executable and not overly simplistic or unsolvable.
- Global Task Level Evaluation: It ensures that the sub-questions in the dialogue cover the expected cross-table retrieval, alignment, and summarization paths, and that each answer aligns with the global task, maintaining logical consistency.

The prompts for sub-question and global task evaluations are as follows:
SUB_QUESTION_EVAL_PROMPT

```
You are a multi-turn dialogue evaluation expert. Your task is to assess the **correctness of the answers** and **the reasonableness of table dependencies** for each sub-question in the multi-turn dialogue plan.

# Input Data Description
1.**Table Overview and Content**: Full content of the tables involved in the questions.
2.**Multi-turn Dialogue Plan**:checkout_list: A detailed list of sub-questions, including the "answer" (expected answer) and "related_tables" (tables involved).

# Task Description
Please analyze each sub-question in the checkout_list as follows:
1.**Table Dependency Reasonableness (table_dependency)**: Verify whether the listed related_tables are necessary to answer the sub-question and whether all tables mentioned are included in related_tables.
2.**Sub-question Difficulty Rating (type_difficulty)**:
-0: Simple query.
-1: Sub-question includes conditions such as filtering, sorting, or aggregating, or it is a standard multi-table query.
- 2: Involves indicator reasoning or computational inference. Indicator reasoning refers to queries where the required metric does not exist directly in the table and must be derived by combining multiple indicators; computational inference involves deducing conclusions by performing calculations on table data.
- 3: Complex multi-table reasoning. Involves reasoning across multiple tables, based on the indicators and calculations described above.
3. **Answer Correctness (correctness)**: Compare the sub-question's answer with the Table Content Overview and assess its accuracy.

# Input Data
## 1. Table Overview and Content (tables separated by "——")
{table_summary}

## 2. Multi-turn Dialogue Plan (Checkout List)
{multi_dialogue_design}

# Output Format
Please output in the following JSON format:
{
  "checkout_list": [
  {
    "id": Sub-question ID (idx),
    "table_dependency": { "reason": "Brief explanation", "value": "True/False/Unknown" },
    "type_difficulty": { "reason": "Brief explanation", "value": 0/1/2/3 },
    "correctness": { "reason": "Explanation of reasoning, specifying which key point or table the original answer is based on and how the inference is made.", "value": "True/False/Unknown" }
  },
  ...
  ]
}
```

TASK_EVAL_PROMPT

You are a multi-turn dialogue evaluation expert. Your task is to assess the **overall quality** and **coverage** of the multi-turn dialogue plan. Assume that the answers to sub-questions and table references are correct, and focus on evaluating the following dimensions.

Input Data Description
1. **Table Overview and Content**: Full content of the tables involved.
2. **Multi-turn Dialogue Plan**: Includes task description (task), design thinking (design_thinking), query path (query_path), and sub-question list (checkout_list).

Evaluation Dimensions
1. **Multi-table Reasoning Difficulty Rating (total_difficulty)**: Calculate the ratio of multi-table reasoning questions to the total number of questions in the sub-question list.
2. **Task Coverage in Multi-turn Dialogue (task_coverage)**:True/False: Whether the information retrieved by the sub-questions fully covers the goal defined in the task.

Input Data

1. Table Overview and Content (tables separated by "——")
{table_summary}

2. Multi-turn Dialogue Plan
{multi_dialogue_design}

Output Format
Please output in the following JSON format:

```
{
  "total_difficulty": { "reason": "Brief explanation", "value": "Number of multi-table questions/Total number of questions" },
  "task_coverage": { "reason": "Brief explanation", "value": "True/False" }
}
```

Manual Review

In the human review stage of the quality control process, experts manually audit the filtered dialogues to correct any remaining errors. This stage is critical for ensuring the accuracy, reliability, and overall quality of the final dataset. The annotation process is structured around several key dimensions, each essential for evaluating dialogue quality:

1. Accuracy of Relevant Table Paths and Dependencies Between Questions and Tables: Experts verify that the correct tables are used throughout the dialogue and that the dependencies between questions and tables are properly maintained. This ensures that each question is supported by the relevant table(s) and that the referenced data sources are both appropriate and accurate.
2. Consistency Between Tasks and Multi-Turn Dialogues: This dimension assesses the alignment between the specified task and the structure of the multi-turn dialogue. Experts ensure that the dialogue progresses logically, with each question and answer contributing to the completion of the overall task. If the dialogue diverges from the intended task or fails to address the task effectively, corrections are made.

3. Clarity of Sub-questions and Adherence to the Specified DAG: Sub-questions are evaluated for clarity, ensuring that each is concise, clear, and unambiguous. Experts also ensure that the sub-questions follow the defined reasoning DAG, which dictates the logical flow and dependencies. If a sub-question is unclear or deviates from the expected structure, it is revised to improve clarity and alignment with the DAG.
4. Factual Correctness of Answers: The factual accuracy of answers is rigorously examined to ensure that each response is supported by the relevant data and is free from errors. Experts confirm that the information in the answers aligns with the provided tables and data, checking for any inaccuracies or unsupported conclusions.

Given the open-ended nature of multi-turn dialogue responses, a rubric-based evaluation protocol is implemented to ensure a structured and consistent approach to manual annotation. Experts are provided with reference answers for key data points, which serve as benchmarks for evaluating the accuracy and completeness of the model-generated responses. The rubric includes specific scoring criteria for critical data and logic, ensuring consistent and transparent dialogue quality assessment.

Scoring point annotation is a critical component of the manual review process. It involves marking the key data and core logic used in the answers to assess the overall quality of the responses. Scoring points focus on the accuracy and relevance of the information provided, as well as the clarity and correctness of the reasoning. For instance, scoring points may include:

- Correctness of comparisons (e.g., comparing traffic volume across regions),
- Accurate use of data points (e.g., ensuring correct numbers from source tables),
- Valid analysis of trends or correlations (e.g., ensuring trends are accurately described and supported by data),
- Appropriateness of derived conclusions (e.g., ensuring conclusions logically follow from the data).

These annotations are vital for evaluating whether model-generated responses meet the required standards and align with the intended task objectives. By carefully reviewing and annotating these key points, experts ensure that the final dataset reflects high-quality dialogue content that accurately captures the reasoning and analysis required for multi-turn tasks.

After manual annotation, the dialogues undergo a final validation process. This ensures that all issues identified during the review are addressed and that the dialogues meet the required quality standards. The revised dialogues are validated for accuracy and consistency, ensuring the dataset is ready for use in subsequent stages of model training, evaluation, and research.

In summary, scoring point annotation is an essential part of the quality control process. It ensures that multi-turn dialogues are accurate, logically consistent, and aligned with task objectives. By combining automated evaluation with expert manual review, we create a robust and reliable dataset suitable for evaluating model performance in complex multi-table reasoning tasks.

Prompt for Skill-Guided Full Dialogue Generation (Rewrite Prompt)

Role: Senior data analyst proficient in multi-turn table analysis and delivering clear responses.

Task: Generate complete multi-turn dialogues that simulate real human interaction, based on the given dialogue skeleton and retrieved skills.

Input Parameters:

- File paths of the original table dataset
- Overall task description
- Dialogue DAG structure (dependency relationships of sub-questions)
- Predefined dialogue skeleton consisting of 2 to 6 turns
- Top 5 relevant skills retrieved for each sub-question

Constraints:

1. Dialogue flow must strictly follow the execution order of sub-questions defined by the dialogue DAG; no skipping or reordering is allowed.
2. Each user query must implicitly imply the need to invoke the most appropriate skill selected from the Top 5 retrieved skills. Do NOT explicitly mention any skill names under any circumstances.
3. Each assistant response must be generated based on table data and accurately present the execution results of the corresponding skill.
4. Retain the core analytical intentions of the original dialogue skeleton entirely and do not deviate from the overall task objectives.
5. Use natural and colloquial language. Avoid rigid imperative expressions and conform to real human interaction habits.
6. Naturally reference analysis results from previous turns to maintain contextual coherence across multi-round interactions.

Output Format: User: [Query of Turn 1] Assistant: [Response of Turn 1] User: [Query of Turn 2] Assistant: [Response of Turn 2] ... User: [Query of Turn N] Assistant: [Response of Turn N]

The 6 domains and 28 subdomains in our benchmark

Domains	Subdomains
Engineering Science	Manufacturing and Industrial Production; Chemical and Materials Engineering; Energy Production and Power Systems; Automotive and Mechanical Engineering; Civil Engineering and Construction; Mining and Resource Extraction; Agricultural Production and Rural Economy
Social Policy Administration	Education and Academic Research; Government Administration and Public Services; Healthcare Systems and Public Health; Social Security and Welfare Programs; Transportation Networks and Infrastructure; Land and Water Resource Management
Business Operations	Supply Chain and Logistics Management; Retail Trade and E-commerce Platforms; Marketing and Sales Management; Human Resources and Corporate Management; International Trade and Commerce
Macroeconomics Statistics	Economic Development and Fiscal Policy; Labor Market and Employment Statistics; Tax Policy and Revenue Management
Financial Services	Banking and Insurance Services; Investment and Capital Markets
Consumer Lifestyle	Tourism and Hospitality Services; Food and Beverage Services; Real Estate and Home Improvement; Entertainment, Sports and Culture; Information Technology and Cybersecurity